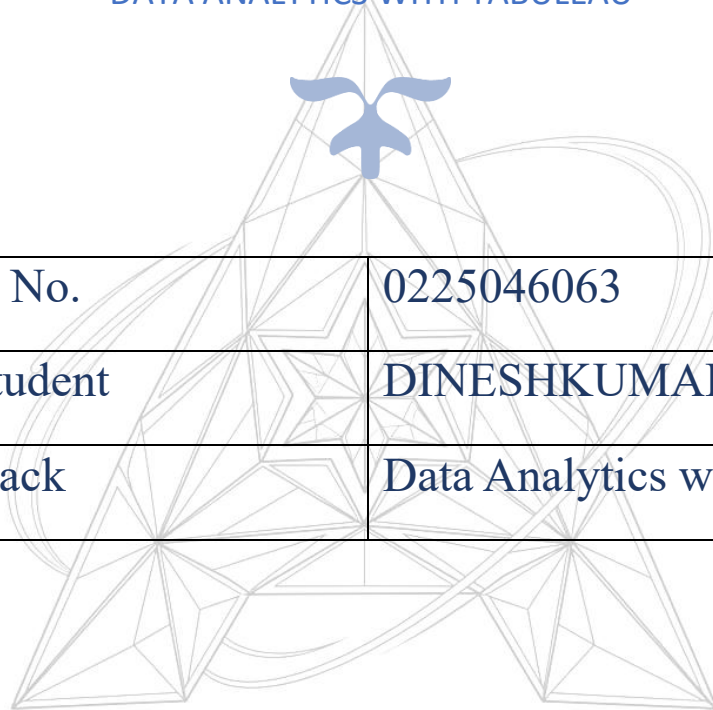


ASSIGNMENT

DATA ANALYTICS WITH TABULEAU



Hall Ticket No.	0225046063
Name of Student	DINESHKUMAR ADITI
Learning track	Data Analytics with Tableau



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[DATE]

[COMPANY NAME]

[Company address]

Data Analytics With Tableau

Assignment

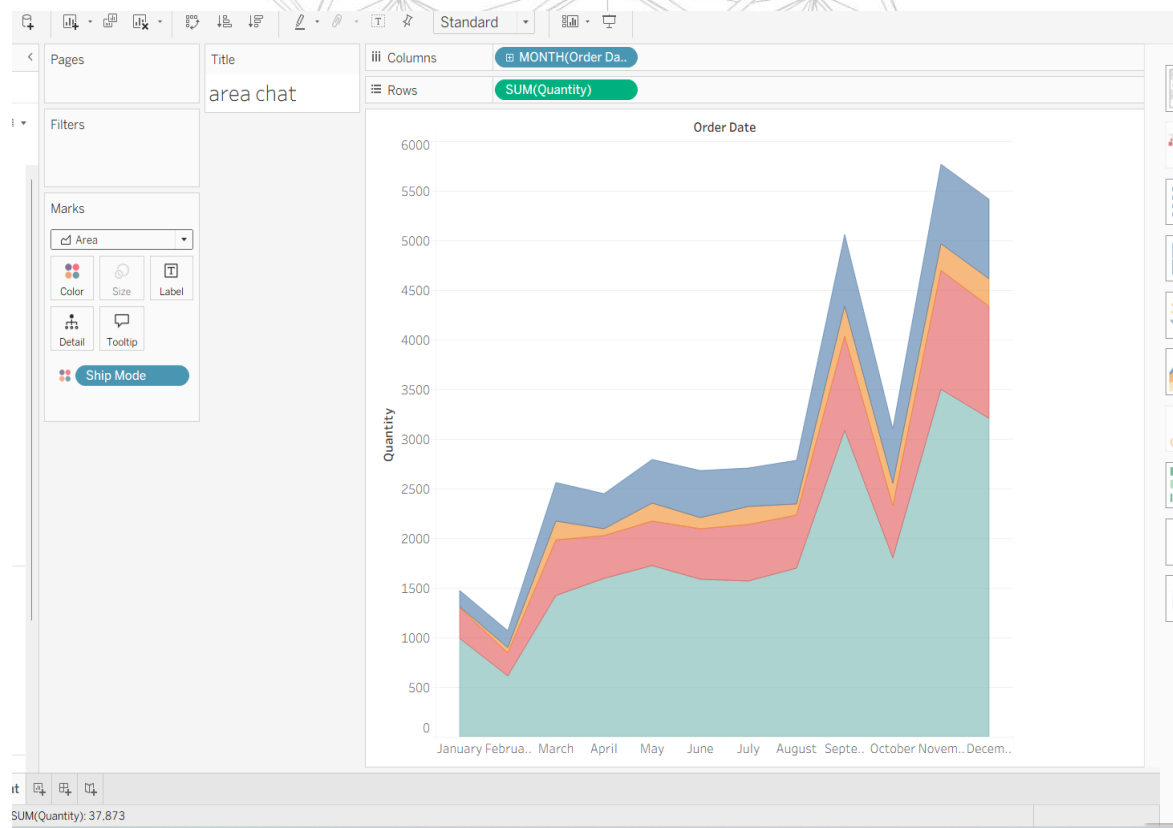
Sheet 01 – Area Chart :

AIM: To learn how to prepare Area Chart.

Procedure:

- 1.To the column shelf, drag the order date from the data panel.
- 2.Right click on year and select month.
- 3.Drag quantity to Row shelf .
- 4.When you drag Ship mode to the Mark card , Select Area as chart type from the drop down menu we create a proper Area chart.

OUTPUT:



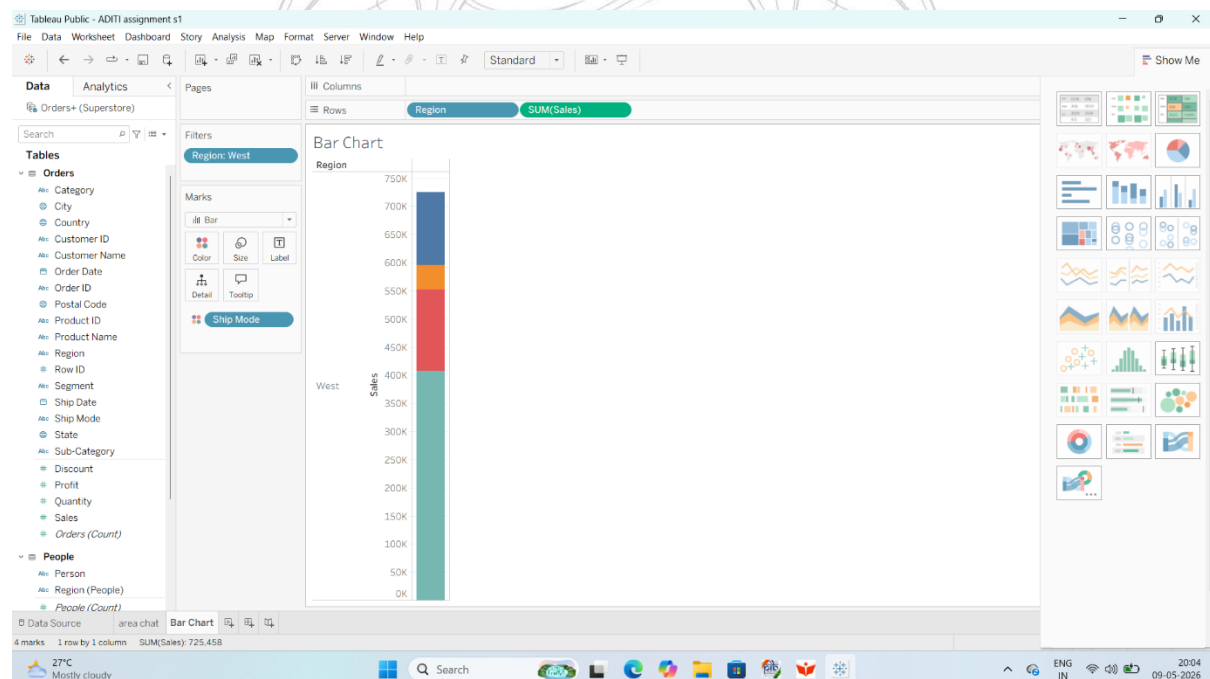
Sheet 02-Bar Chart:

AIM: To learn how to create a Bar Chart.

Procedure:

1. To the row shelf drag sales.
 2. The Tableau aggregates sales as sum(sales), tableau will show a line chart.
 3. Click the drop down menu.
 4. Select Bar.
 5. To add colour ,drag Ship mode to colour.
 6. Drag Region to Row shelf.
 7. Uncheck all except West from the dialog box.
- Your BAR CHART is ready.

OUTPUT:



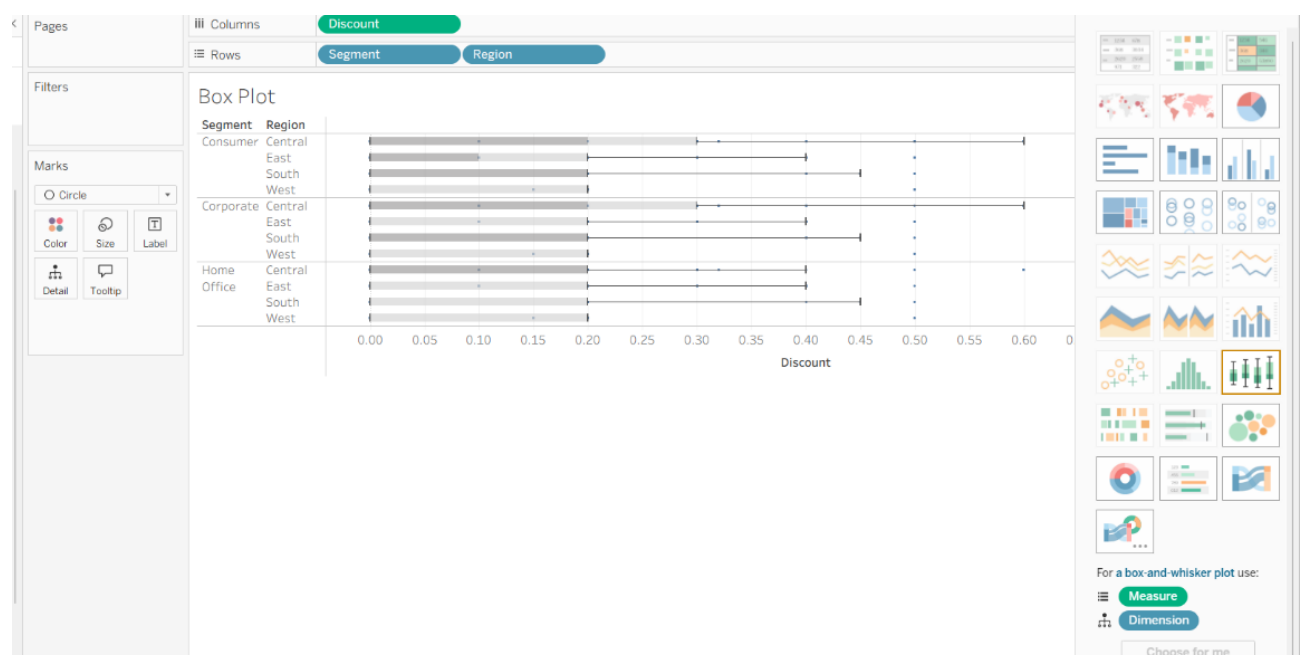
Sheet 03-Box Plot:

AIM: To learn how to make a Box Plot .

Procedure:

1. Drag Segment to Columns shelf.
2. Drag Discount to Rows shelf.
3. To change Aggregation to Average , Click Discount on Rows shelf and now change Aggregation to Average.
4. Drag Region to Column shelf on right side of Segment.
5. Click Show Me in tool bar and select Box-and -Whisker plot.
6. Drag Region from Mark card back to Column shell.
7. Go to Analysis menu and uncheck Aggregate measures.
8. To get a horizontal view click Swap Rows and Columns button.

OUTPUT:



Sheet 04 – Gantt Chart:

AIM: To learn how to make Gantt Chart.

Procedure:

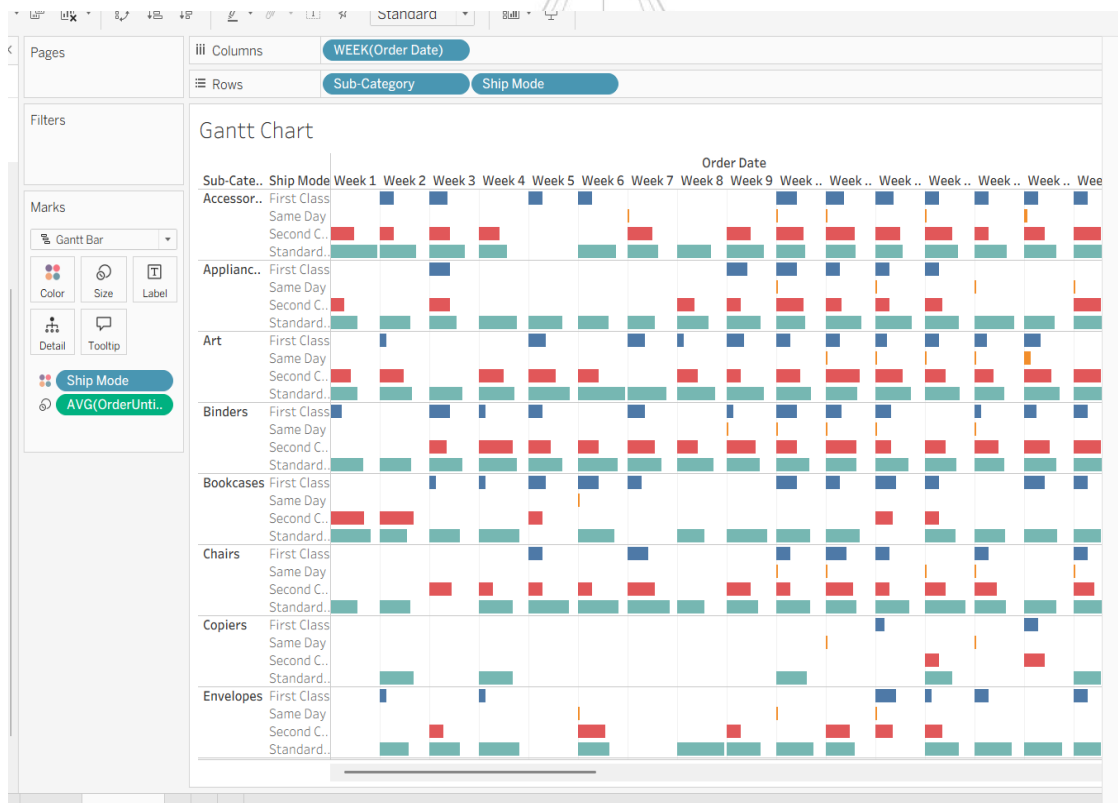
1. Drag Order Date from Dimensions.
2. Drop it on the Columns Shelf.
3. Tableau automatically shows YEAR(Order Date).
4. Click drop-down arrow on YEAR(Order Date) in Columns shelf.
5. Select: More → Week Number Now columns display weekly timeline.
6. Drag Sub-Category to Rows Shelf.
7. Drag Ship Mode to Rows Shelf and place it right side of Sub-Category. Now rows show:
 - Furniture → First Class / Standard Class etc
 - Office Supplies → etc.
8. In Marks card click dropdown.
9. Select Gantt Bar Now bars appear across timeline. We need shipping delay days.
 - Click Analysis → Create Calculated Field
 - Name: OrderUntilShip
 - Formula: DATEDIFF('day',[Order Date],[Ship Date])
 - Click OK
10. Drag OrderUntilShip to Size on Marks card. Bars become longer based on shipping days

11. Change SUM to AVG, Right-click SUM(OrderUntilShip) on Marks card.

12. Select: Measure → Average Now average shipping delay is shown.

13. Drag Ship Mode to Color in Marks card. Now each ship mode gets different color.

OUTPUT:



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Sheet 05 – Heat Map:

AIM: To learn how to make Heat Map.

Procedure:

1. Drag Segment from Dimensions.

2. Drop it on Columns Shelf.

You will see:

- Consumer
- Corporate
- Home Office

3. Drag Region to Rows Shelf

4. Drag Sub-Category to Rows shelf and place it to the right of Region.

5. To Add Profit to Color, Drag Profit measure and Drop it on Color in Marks card.

Now

- High Profit = Blue / Dark
- Loss = Orange

6. Click Color on Marks card.

7. Under Border, choose Medium Gray, This makes each box clearer.

8. To Change Color Palette, On Profit color legend click dropdown arrow and Select Edit Colors In dialog box

: • Palette = Red-Black Diverging

- Check Use Full Color Range
- Click Apply
- Click OK

Now:

- Profit = Black / Dark
- Loss = Red

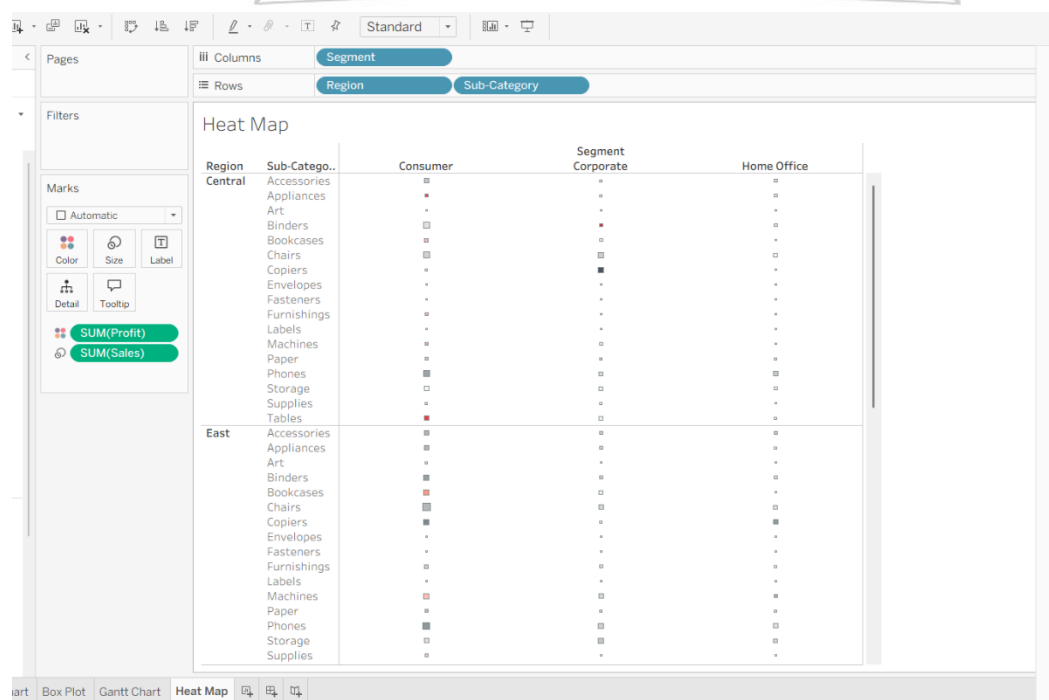
9. Drag Sales measure.

10. Drop it on Size in Marks card. Now box size represents Sales amount.

- Large box = High Sales
- Small box = Low Sales

11. To Increase Box Size, Click Size on Marks card and Move slider to right until boxes fit properly.

OUTPUT:



Sheet 06 – Histogram:

AIM: To create a Histogram.

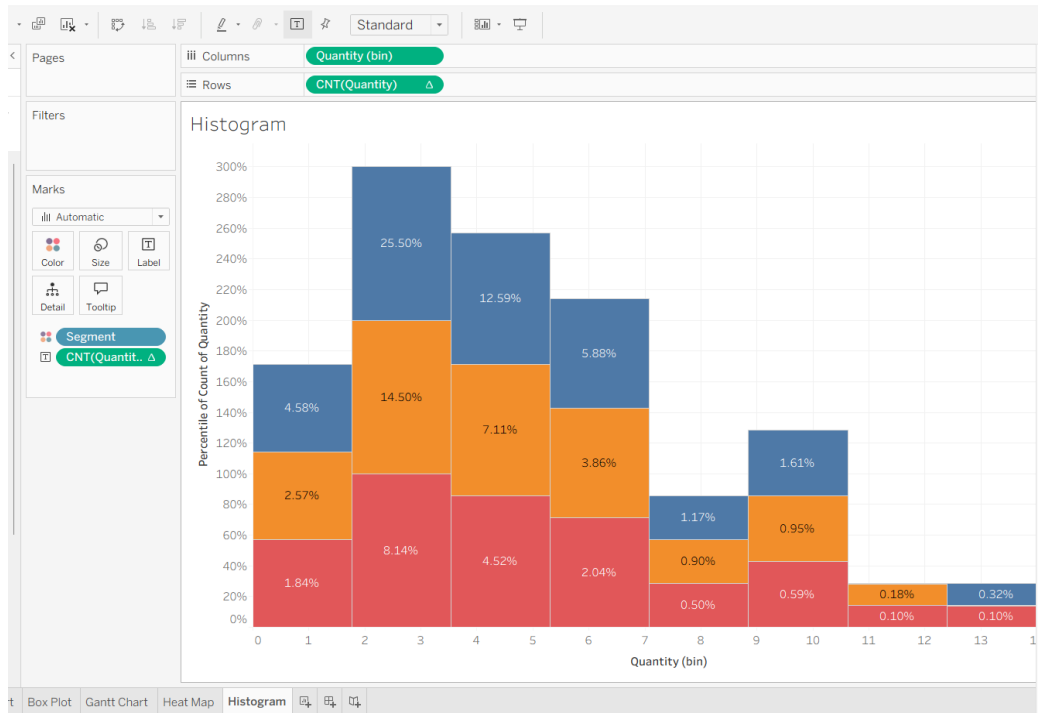
Procedure:

1. From Measures, drag Quantity.
 2. Drop it on Columns Shelf.
 3. Click Show Me (top right corner).
 4. Select Histogram Tableau Automatically Does:
 - Creates Quantity (bin) on Columns
 - Moves Quantity to Rows
 - Changes aggregation to CNT(Quantity)
 5. It shows ,X-Axis: Quantity Bins
Y-Axis: Count of Orders
- Example:
- Around 4800 orders had quantity = 2
 - Around 2400 orders had quantity = 4
6. Drag Segment from Dimensions
 7. Drop it on Color in Marks card Now bars become stacked by:
 - Consumer
 - Corporate
 - Home Office .
 8. Drag CNT(Quantity) from Rows shelf to Label on Marks card (Using Ctrl copies field.)

9. Right-click CNT(Quantity) on Marks card

10. Select: Quick Table Calculation → Percent of Total Now labels show percentages.

OUTPUT:



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Sheet 07-Line Chart:

AIM: To Create a Line Chart.

Procedure:

1. Drag Ship Date from Dimensions.
2. Drop it on Columns Shelf. Tableau shows: YEAR(Ship Date)
3. Drag Sales measure.
4. Drop it on Rows Shelf Tableau creates a line chart.
5. Drag Profit measure.
6. Drop it on Rows shelf, right side of Sales.

Now Tableau creates two separate axes:

- Sales Axis
- Profit Axis

7. Drag SUM(Profit) from Rows shelf.
8. Drop it directly on the Sales Axis. (You will see two pale green lines.)

Release mouse. Now Sales and Profit share same axis.

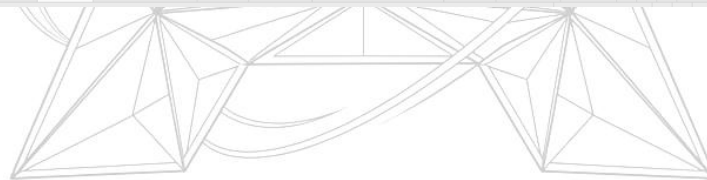
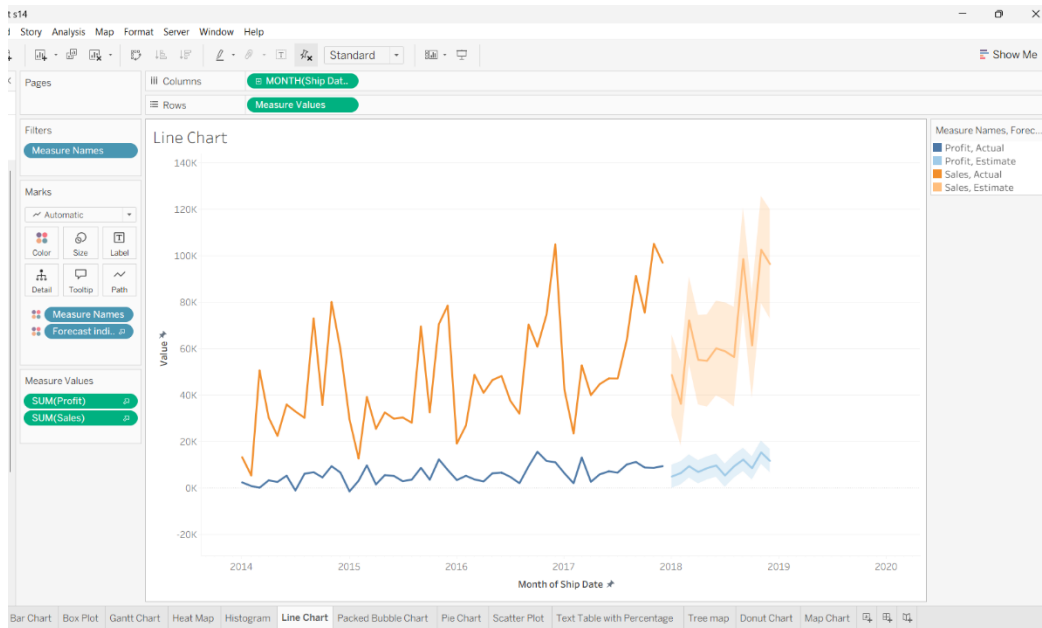
9. Click dropdown on YEAR(Ship Date) in Columns shelf.
10. Select: Month (from lower continuous section) Now chart becomes monthly trend line across years.

- Higher values near year-end
- Repeating yearly spikes
- Seasonal business growth This is called Seasonality.

11. Add Forecast, Click Analytics Pane and Drag Forecast.

12.Drop into chart area Tableau automatically predicts future trend.

OUTPUT:



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Sheet 08-Packed Bubble Chart:

AIM: To create a Packed Bubble Chart.

Procedure:

1. Drag Category from Dimensions
2. Drop it on Columns Shelf You will see:
 - Furniture
 - Office Supplies
 - Technology
3. To Add Sales, Drag Sales measure and Drop it on Rows Shelf Tableau creates a Bar Chart by default.
4. To Change to Packed Bubbles, Click Show Me (top right) and Select Packed Bubbles Now Tableau creates bubbles for:
 - Furniture
 - Office Supplies
 - Technology Bubble size = Sales
5. To Add Region to Detail, Drag Region and Drop it on Detail in Marks card Now each Category splits into:
 - East
 - West
 - Central
 - South More bubbles appear
6. Add Profit to Color: Drag Profit and Drop it on Color. Now bubble color represents Profit.

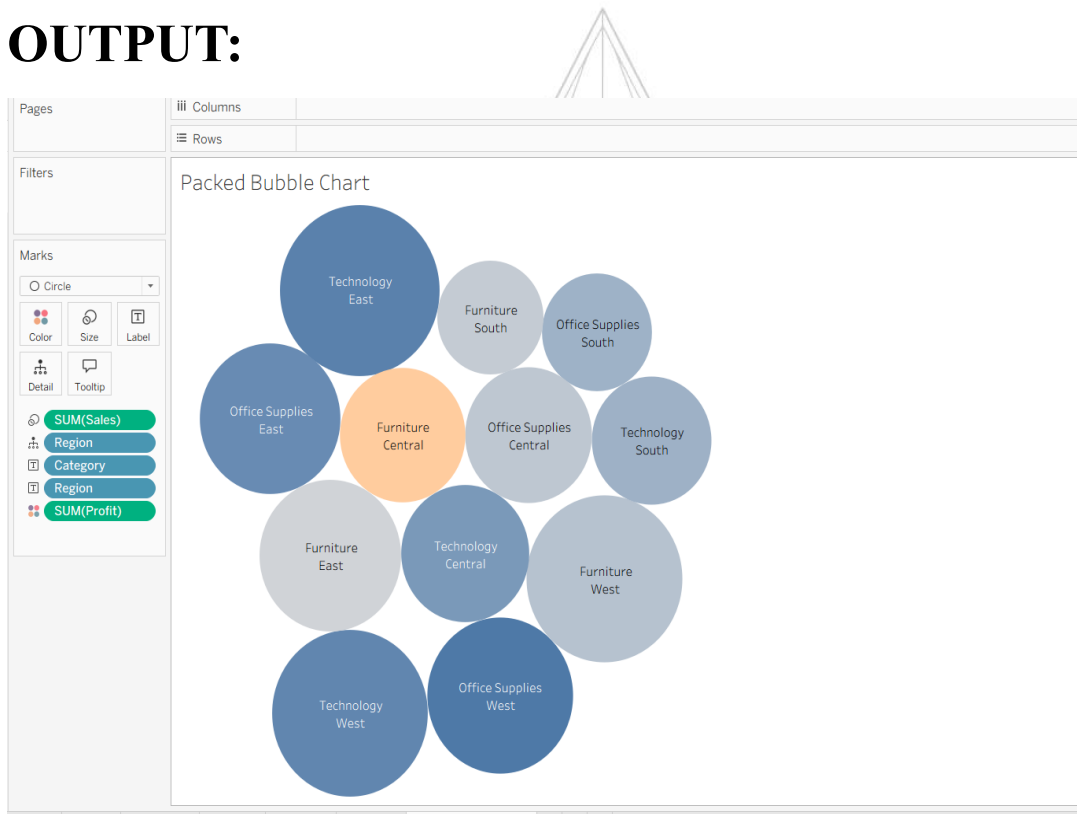
Example: • Dark Blue = High Profit

• Light Blue = Lower Profit

• Orange/Red (if palette changed) = Loss

7. Add Region to Label. Drag Region and Drop it on Label
Now each bubble displays region name.

OUTPUT:



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Sheet 09- Pie Chart:

AIM: To create Pie Chart Diagram.

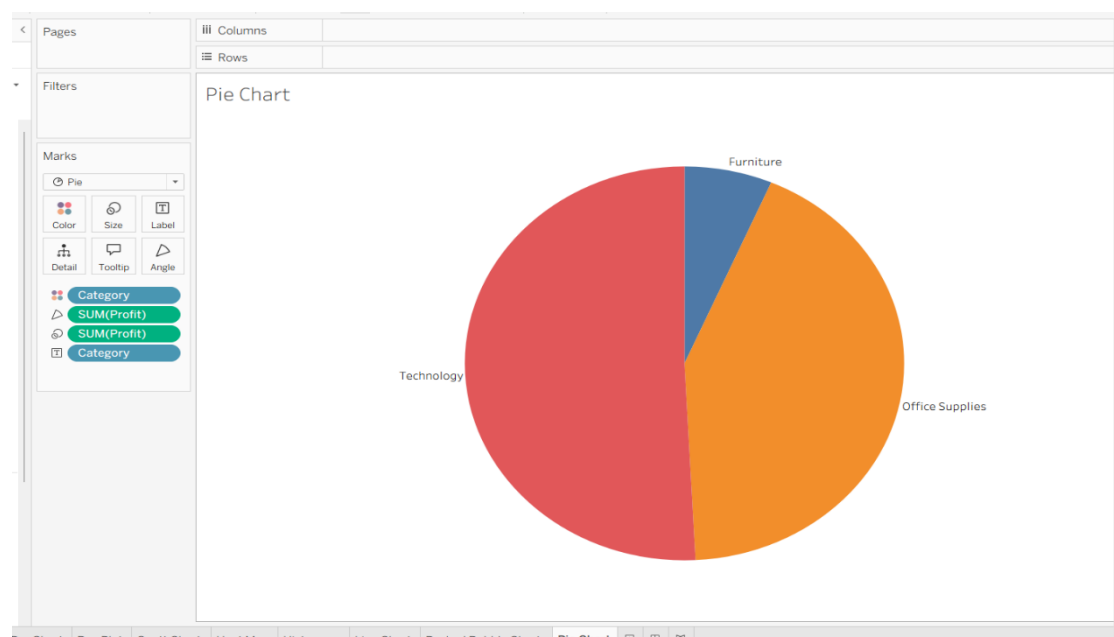
Procedure:

1. Drag Profit measure, and Drop it on Columns Shelf
2. Drag Category dimension, and Drop it on Rows Shelf
Tableau creates a Bar Chart by default.
3. Click Show Me, and Select Pie Chart
4. To Increase Pie Chart Size Option, Click Fit menu (toolbar) and Select Entire View Option
5. Drag Category, and Drop it on Label in Marks card

Now chart displays:

- Furniture
- Office Supplies
- Technology

OUTPUT:



Sheet 10 – Scatter Plot:

AIM: To create scatter plot.

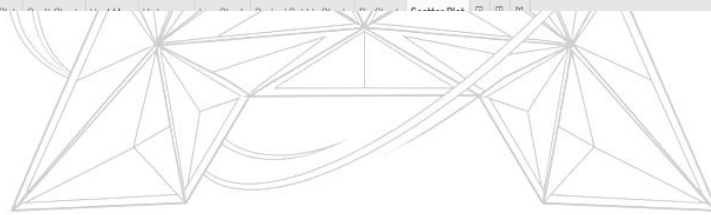
Procedure:

1. Drag Sales measure, and Drop it on Columns Shelf
Tableau creates horizontal axis.
 2. Drag Profit measure, and Drop it on Rows.
 3. Drag Category dimension, and Drop it on Color in Marks card Now marks split into:
 - Furniture
 - Office Supplies
 - Technology Each category gets different color.
 4. Drag Region, and Drop it on Detail in Marks card. Now each Category is divided by:
 - Central
 - East
 - South
 - West
- Total marks = Category × Region
5. Click Analytics Pane, Drag Trend Line and Drop it on char.
 6. Select: Linear

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OUTPUT:



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Sheet 11 -Text Table With Percentages:

AIM: To create Text Table With Percentage.

Procedure:

1. Drag Order Date dimension, and Drop it on Columns Shelf
Tableau displays: YEAR(Order Date)

2. Drag Sub-Category, and Drop it on Rows Shelf Now a table structure is created.

Rows = Sub-Categories Columns = Years

3. Drag Sales, and Drop it on Text in Marks card. Now each cell shows: SUM(Sales) for each year and sub-category.

4. Drag Region, and Drop it to the left side of Sub-Category in Rows shelf .Now rows become:

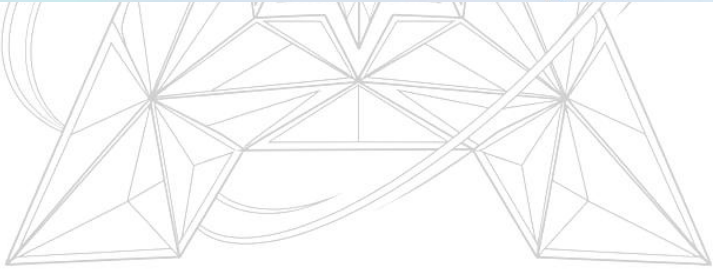
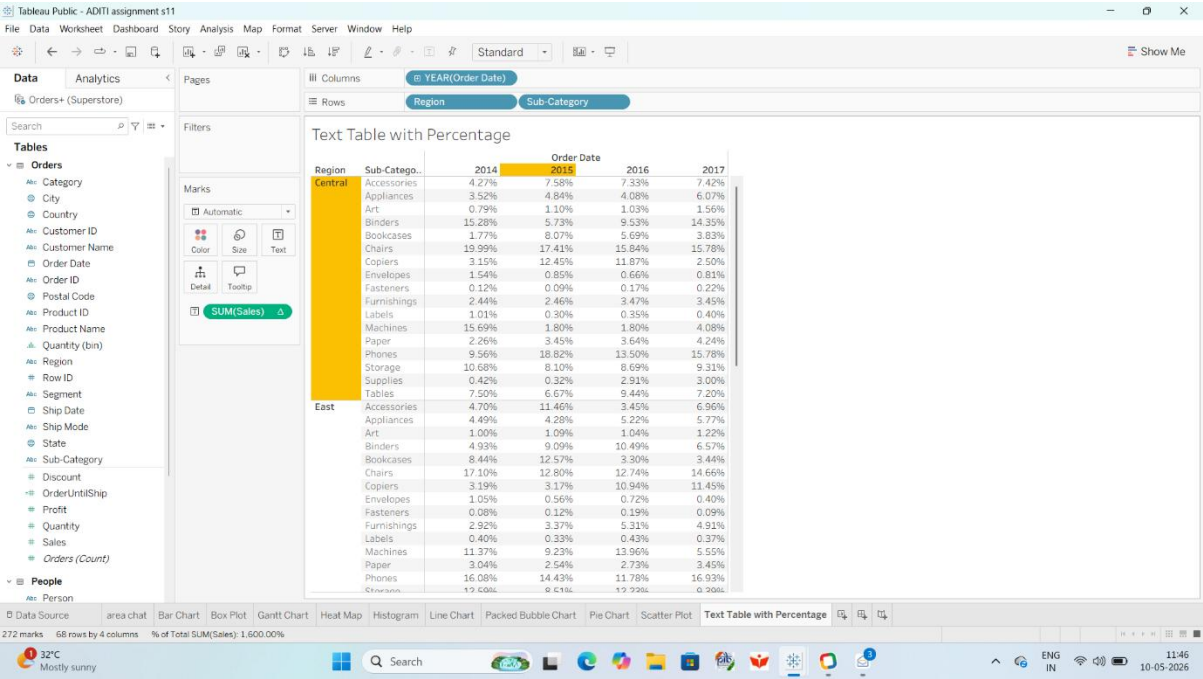
- Central → Chairs
- Central → Phones
- East → Chairs
- East → Phones etc.

5. To Convert Sales to Percentages, Right-click SUM(Sales) on Marks card . Select: Add Table Calculation

6. Configure Table Calculation In Dialog Box: Calculation Type: Percent of Total Compute Using: Pane (Down)

7. click OK

OUTPUT:



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Sheet 12- Tree Map

AIM: To learn how to create a Tree Map.

Procedure:

1. Drag Sub-Category dimension , Drop it on Columns Shelf
2. Drag Sales measure , Drop it on Rows Shelf Tableau creates a Bar Chart by default.
3. Click Show Me , Select Treemap

Now Tableau creates rectangles for each Sub-Category.

Default Treemap Meaning Feature Represents Rectangle Size
Sales Rectangle Color Sales

Example: • Large Dark Box = High Sales

• Small Light Box = Low Sales

4. Drag Ship Mode , Drop it on Color in Marks card. Now rectangles are colored by:

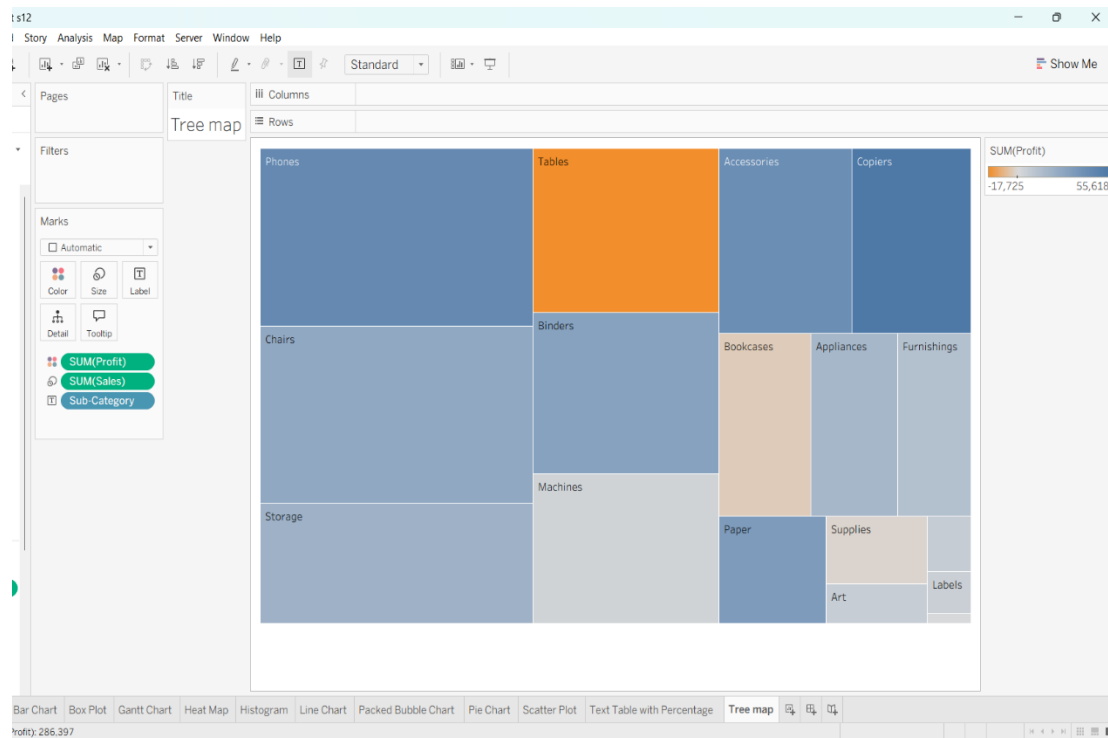
- First Class
- Second Class
- Standard Class
- Same Day

Sales still controls size.

5. Click Undo to remove Ship Mode, Drag Profit . Drop it on Color Now:

- Size = Sales
- Color = Profit

Output:



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Sheet 13 -Donut Chart:

AIM: To create a Donut Chart.

Procedure:

1.: To create Calculated field

- Go to the Analysis menu.
- Select Create Calculated Field.
- Name the field as Zero.
- In the formula area, type: 0
- Click OK.

2: Drag Zero to Row ,From the Measures section, drag Zero to the Rows shelf

.3: Convert into Pie Chart

- In the Marks card, click Automatic.
- Select Pie.
- From the toolbar, choose Fit → Entire View.

4: Increase Pie Size

- In the Marks card, click Size
- Increase the size slider.
- This creates a larger pie chart.

5 : Create Second Pie for Donut Hole.

- drag Zero from Measures to the Rows shelf.
- Place it on the right side of the first SUM(Zero).

Now two pie charts appear.

6:Create Dual Axis

- Right-click on the second SUM(Zero) in Rows.
- Select Dual Axis. Now both pie charts overlap.

7: Create the Inner Circle

- In the Marks card, click the second SUM(Zero).
- Click Size and reduce the size slightly.
- Click Color.
- Select White color. This creates the donut hole effect.

8: Add Region Colors

- Click the first SUM(Zero) Marks card.
- Drag Region to Color. Now each region gets different colors.

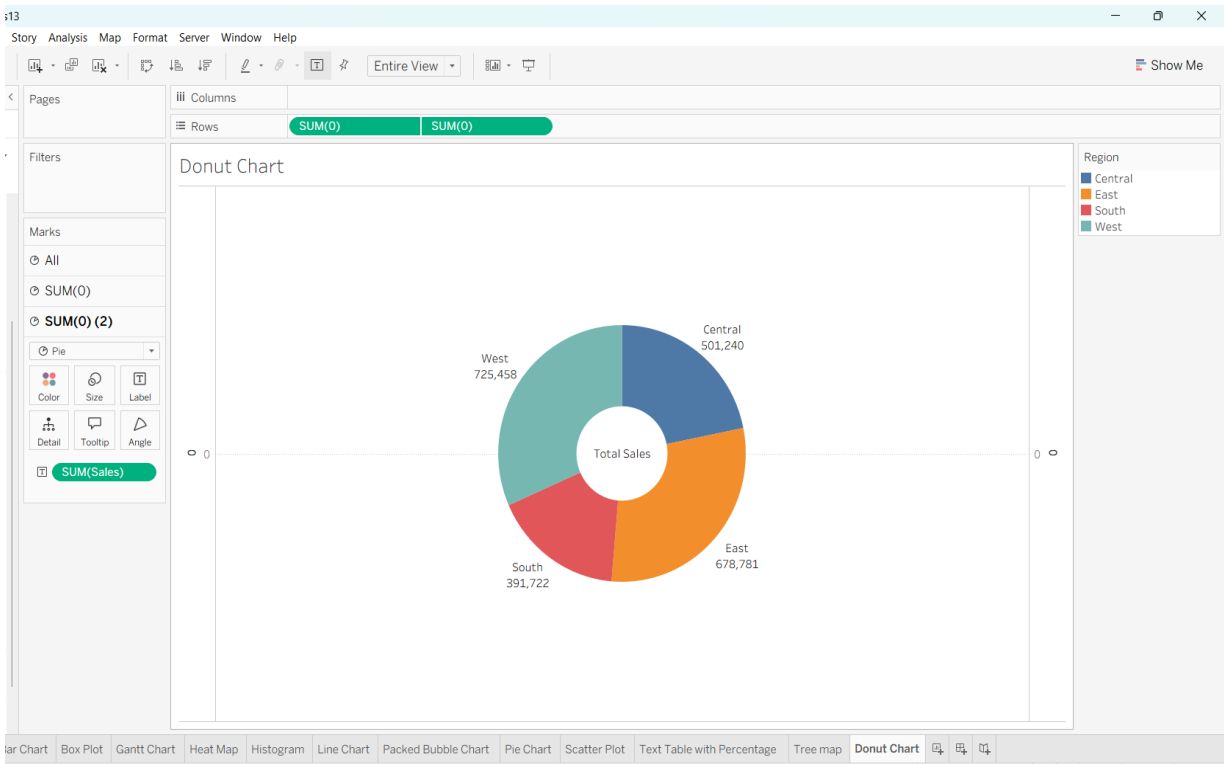
9: Add Labels

- Drag Region to Label on the Marks card.
- Drag Sales to Label.
- Drag Sales to Angle. The pie slices now display sales values and adjust according to sales.

10: Add Total Sales in Center

- Click the second SUM(Zero) Marks card.
- Drag Sales to Label.
- Click Label → Text.
- Click the three dots (...) beside text editing. Type: Total Sales
- Click OK. Now the center of the donut chart displays Total Sales.

OUTPUT:



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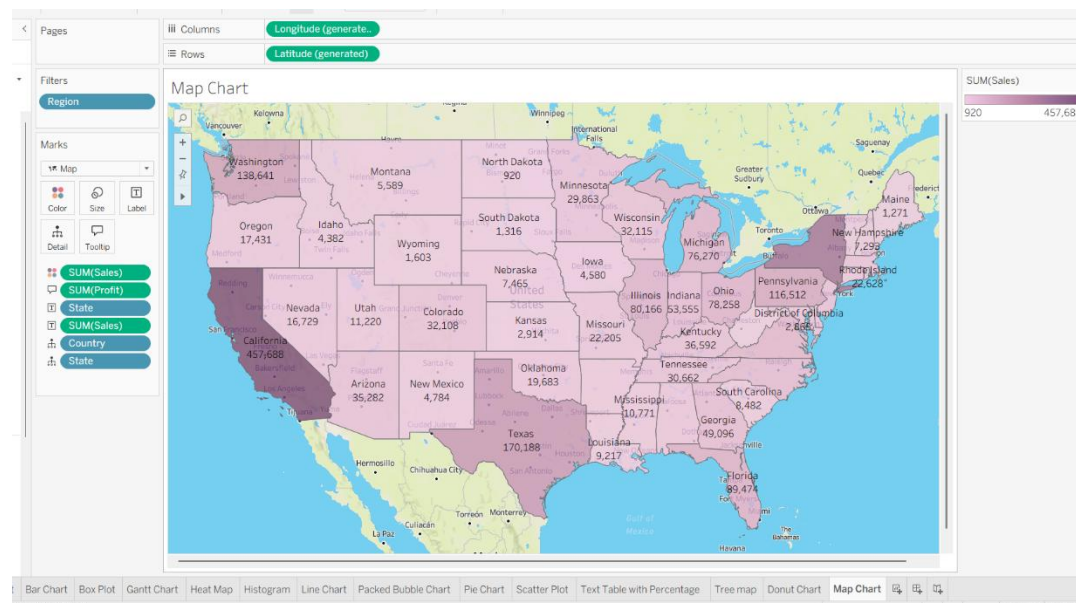
Sheet 14 – Map Chart

AIM: To learn to create a Map Chart.

Procedure :

- 1.From Data pane drag Country to the workspace.
- 2.Drag Sales to Colour on Markcard.
- 3.In the Mark card ,Click Automatic.
4. Select Map.
5. Bring States to workplace
- 6.Click Colour on Mark card , Select Edit Colour .
7. Choose purple palette.
8. Drag Region to Filter shelf.
9. Drag Profit to Tooltip.
10. Go to Map layer and select Map.
- 11.Adjust Backgroud style to Out style.

OUTPUT:



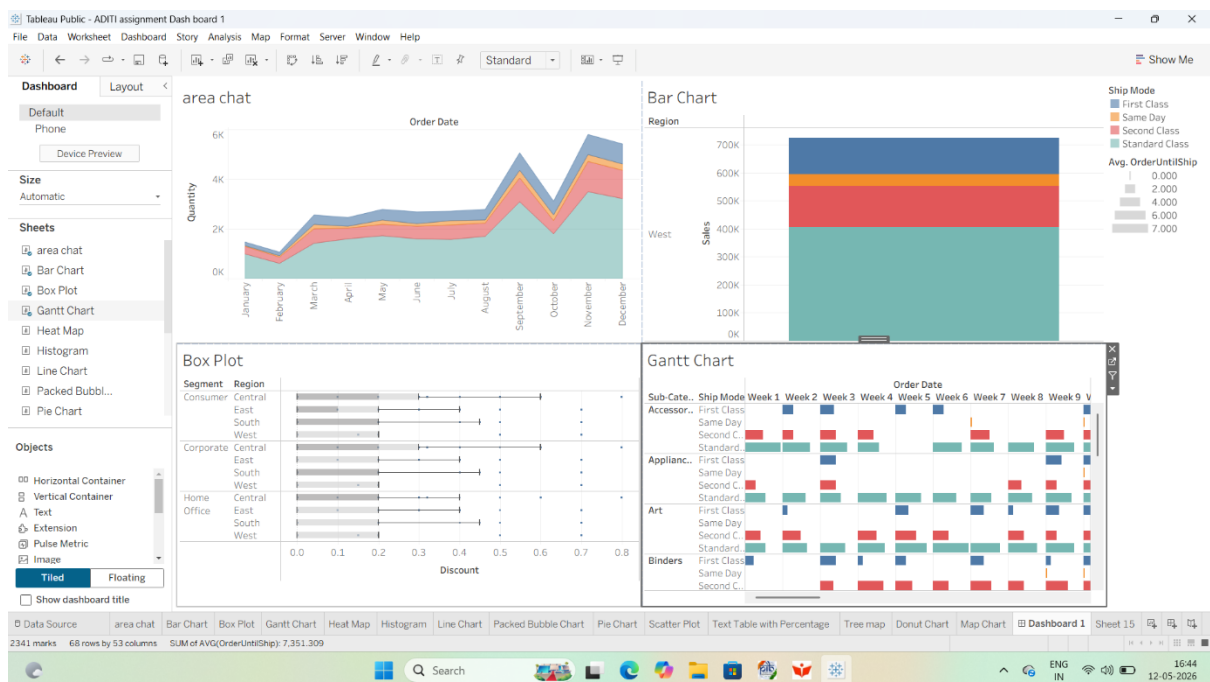
DASHBOARD

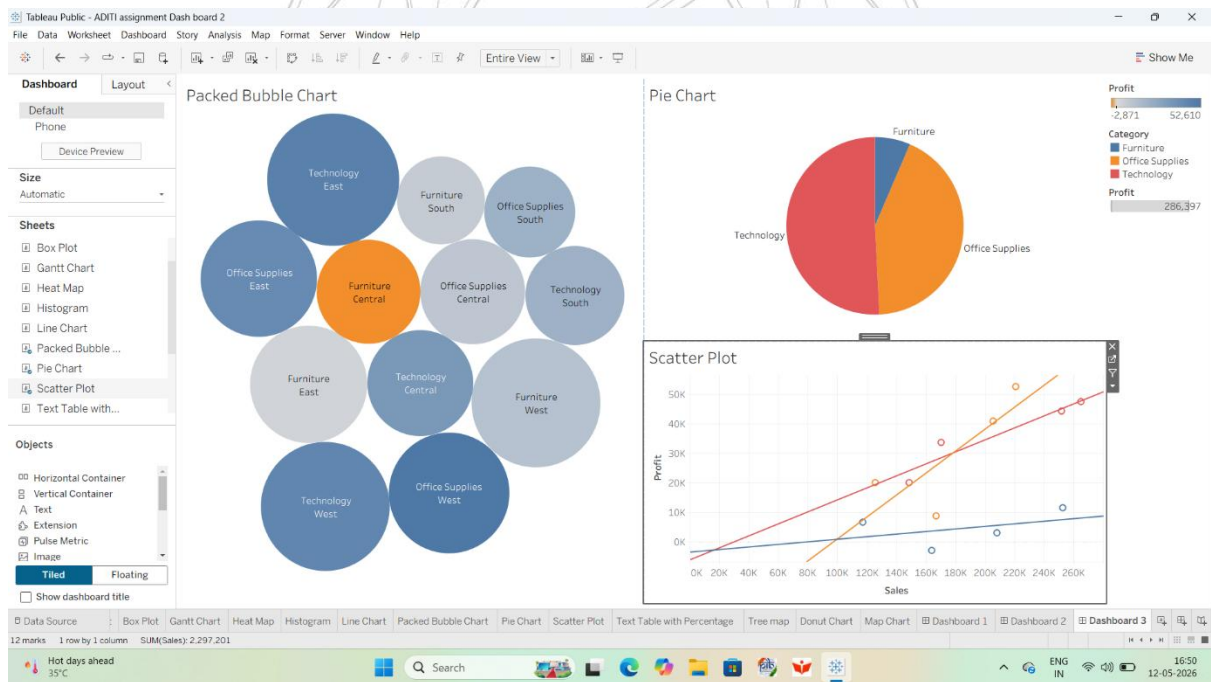
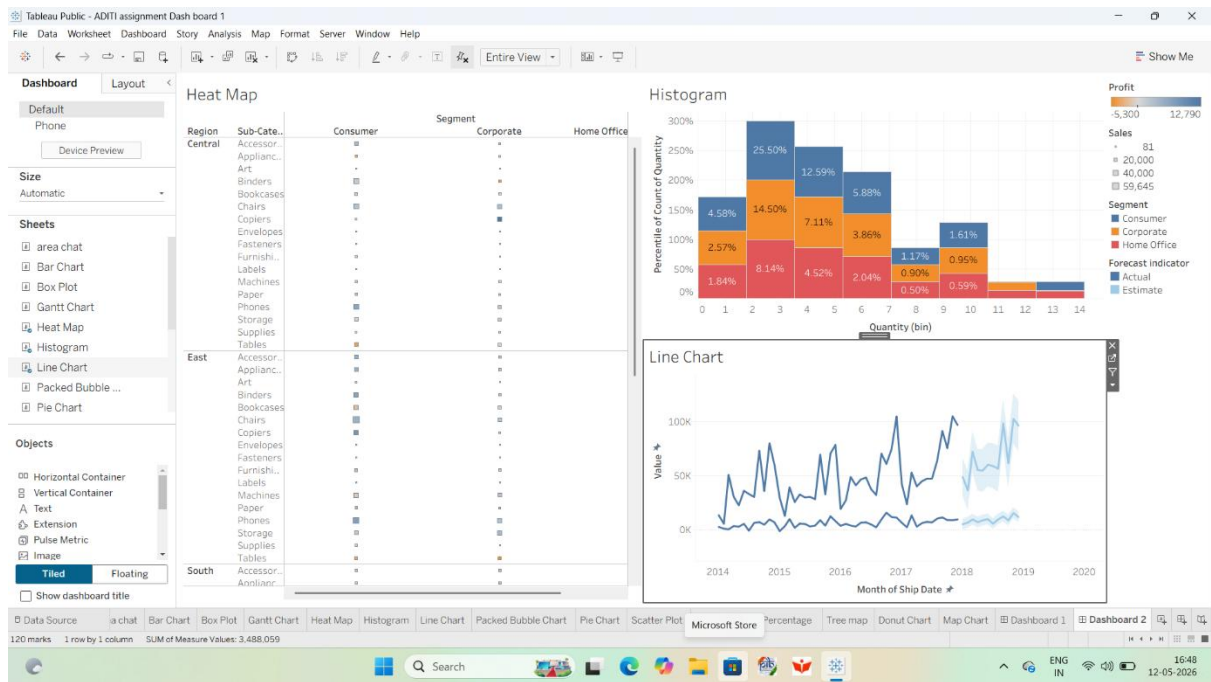
AIM : To create a Dashborad.

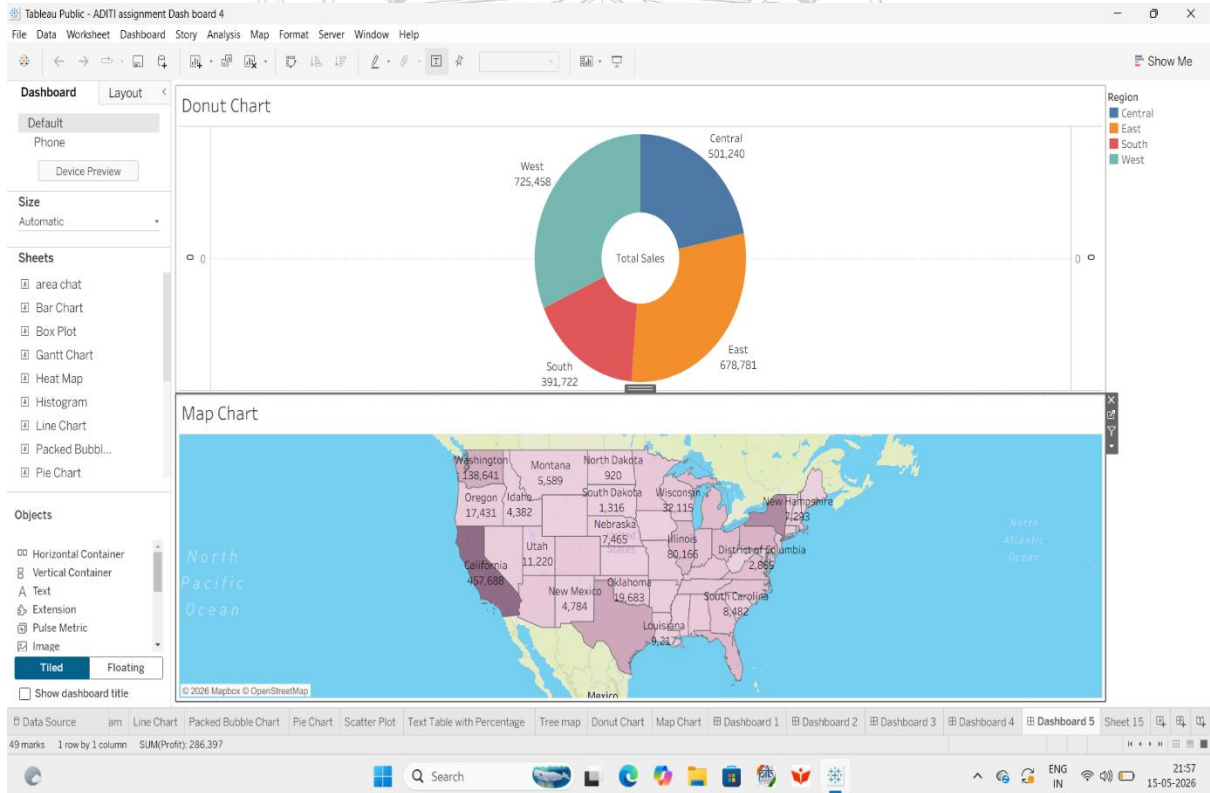
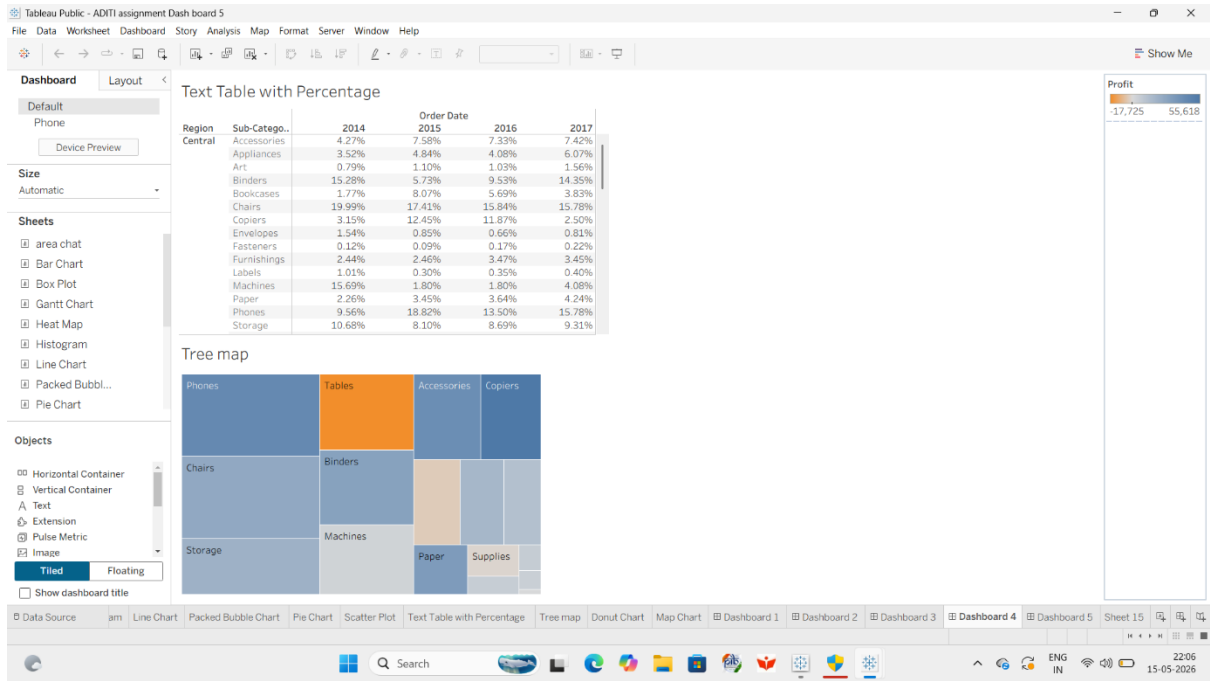
Procedure;

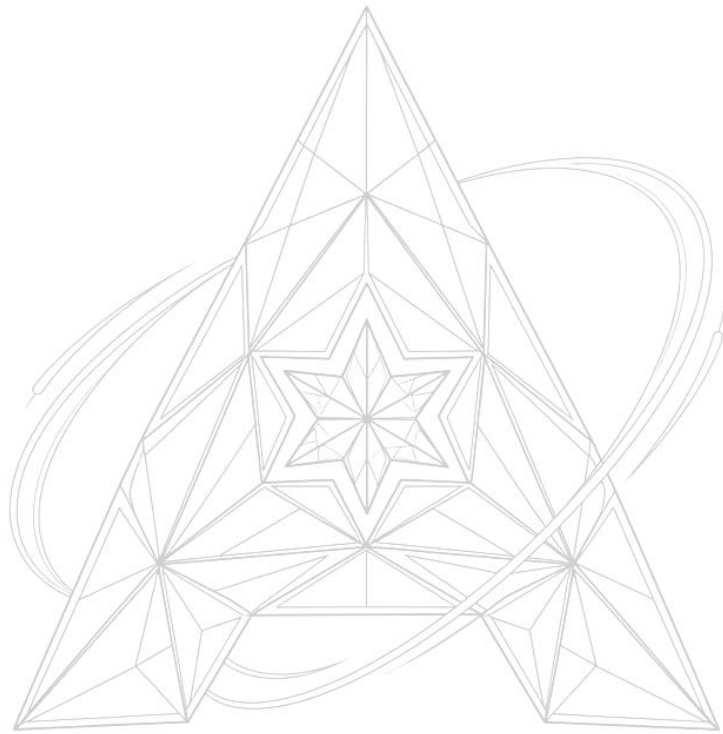
1. Click on DASHBOARD top above.
2. Select New Dashboard and a black Dashboard will appear on screen
3. Adjust the size by clicking on Automatic option. A full screen Dashboard will appear.
4. On the left side Assignment list will be given to add it on dashboard. You can collage pictures as per your wish
5. pictures can be easily adjusted and moved from one place to another.

OUTPUT:









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